

PowerShell vs Linux Commands

File & Directory Management

PowerShell Command	Linux Command	Uses	Example
Get-ChildItem	ls	List files and directories	Example: <code>Get-ChildItem</code> <code>ls</code>
Get-ChildItem - Recurse	find .	Recursive file listing	Example: <code>Get-ChildItem - Recurse</code> <code>find .</code>
Get-ChildItem -Filter *.log	ls *.log	List filtered files	Example: <code>Get-ChildItem -Filter *.log</code> <code>ls *.log</code>
New-Item file.txt	touch file.txt	Create a new file	Example: <code>New-Item file.txt</code> <code>touch file.txt</code>
Remove-Item file.txt	rm file.txt	Delete a file	Example: <code>Remove-Item file.txt</code> <code>rm file.txt</code>
Copy-Item file.txt C:\	cp file.txt /tmp	Copy file	Example: <code>Copy-Item file.txt C:\</code> <code>cp file.txt /tmp</code>
Move-Item file.txt C:\	mv file.txt /tmp	Move file	Example: <code>Move-Item file.txt C:\</code> <code>mv file.txt /tmp</code>
Rename-Item old.txt new.txt	mv old.txt new.txt	Rename file	Example: <code>Rename-Item old.txt new.txt</code> <code>mv old.txt new.txt</code>
Get-Content file.txt	cat file.txt	View file contents	Example: <code>Get-Content file.txt</code> <code>cat file.txt</code>
Get-Content file.txt - Tail 5	tail -n 5 file.txt	View last 5 lines	Example: <code>Get-Content file.txt -Tail 5</code> <code>tail -n 5 file.txt</code>
Set-Content file.txt "Hello"	echo "Hello" > file.txt	Write to file	Example: <code>Set-Content file.txt "Hello"</code> <code>echo "Hello" > file.txt</code>
Add-Content file.txt "World"	echo "World" >> file.txt	Append to file	Example: <code>Add-Content file.txt "World"</code> <code>echo "World" >> file.txt</code>

 Process Management

PowerShell Command	Linux Command	Uses	Example
Get-Process	ps aux	List running processes	Example: <code>Get-Process</code> <code>ps aux</code>
Stop-Process -Name notepad	killall notepad	Stop a process by name	Example: <code>Stop-Process -Name notepad</code> <code>killall notepad</code>
Get-Process Sort-Object CPU	ps aux --sort=-%cpu	Sort processes by CPU usage	Example: <code>Get-Process Sort-Object CPU -Descending</code> <code>ps aux --sort=-%cpu</code>
Get-Process Sort-Object WS	ps aux --sort=-%mem	Sort processes by memory	Example: <code>Get-Process Sort-Object WS -Descending</code> <code>ps aux --sort=-%mem</code>
Get-Process notepad -ErrorAction SilentlyContinue	pgrep notepad	Check if process is running	Example: <code>Get-Process notepad -ErrorAction SilentlyContinue</code> <code>pgrep notepad</code>

 Services & System Control

PowerShell Command	Linux Command	Uses	Example
Get-Service	systemctl list-units --type=service	List services	Example: <code>Get-Service</code> <code>systemctl list-units --type=service</code>
Start-Service spooler	systemctl start cups	Start a service	Example: <code>Start-Service spooler</code> <code>systemctl start cups</code>
Stop-Service spooler	systemctl stop cups	Stop a service	Example: <code>Stop-Service spooler</code> <code>systemctl stop cups</code>
Restart-Service spooler	systemctl restart cups	Restart a service	Example: <code>Restart-Service spooler</code> <code>systemctl restart cups</code>
Restart-Computer	reboot	Restart system	Example: <code>Restart-Computer</code> <code>reboot</code>
Stop-Computer	shutdown now	Shutdown system	Example: <code>Stop-Computer</code> <code>shutdown now</code>

PowerShell Command	Linux Command	Uses	Example
Stop-Computer - Force	poweroff	Force shutdown	Example: Stop-Computer -Force poweroff

 Networking

PowerShell Command	Linux Command	Uses	Example
Test-Connection google.com	ping google.com	Ping a host	Example: Test-Connection google.com ping google.com
Invoke-WebRequest http://example.com	curl http://example.com	Download web content	Example: Invoke-WebRequest http://example.com curl http://example.com
Get-NetIPAddress	ip addr show	Show IP addresses	Example: Get-NetIPAddress ip addr show
Get-NetTCPConnection	ss -tuln	Show TCP connections	Example: Get-NetTCPConnection ss -tuln

 User Management

PowerShell Command	Linux Command	Uses	Example
Get-LocalUser	cat /etc/passwd	List users	Example: Get-LocalUser cat /etc/passwd
New-LocalUser testuser	useradd testuser	Create a new user	Example: New-LocalUser testuser useradd testuser
Remove-LocalUser testuser	userdel testuser	Delete a user	Example: Remove-LocalUser testuser userdel testuser
Set-LocalUser testuser - Password 'Pass@123'	passwd testuser	Set user password	Example: Set-LocalUser testuser - Password 'Pass@123' passwd testuser

 Disk & File System

PowerShell Command	Linux Command	Uses	Example
Get-PSDrive	df -h	Check disk space	Example: Get-PSDrive df -h

PowerShell Command	Linux Command	Uses	Example
Get-Item C:\file.txt	stat file.txt	Show file details	Example: <code>Get-Item C:\file.txt</code> <code>stat file.txt</code>
Get-Volume	lsblk	List drives/partitions	Example: <code>Get-Volume</code> <code>lsblk</code>
Mount-DiskImage -ImagePath C:\disk.iso	mount -o loop disk.iso /mnt	Mount ISO	Example: <code>Mount-DiskImage -ImagePath C:\disk.iso</code> <code>mount -o loop disk.iso /mnt</code>
Dismount-DiskImage -ImagePath C:\disk.iso	umount /mnt	Unmount ISO	Example: <code>Dismount-DiskImage -ImagePath C:\disk.iso</code> <code>umount /mnt</code>

 Permissions & Security

PowerShell Command	Linux Command	Uses	Example
Get-Acl file.txt	ls -l	View file permissions	Example: <code>Get-Acl file.txt</code> <code>ls -l file.txt</code>
Set-Acl file.txt	chmod 644 file.txt	Change file permissions	Example: <code>Set-Acl file.txt</code> <code>chmod 644 file.txt</code>
New-ItemProperty	chown user:user file	Change ownership	Example: <code>New-ItemProperty -Path file.txt</code> <code>chown user:user file</code>

 Logging & Monitoring

PowerShell Command	Linux Command	Uses	Example
Get-EventLog -LogName System -Newest 10	journalctl -n 10	View system logs	Example: <code>Get-EventLog -LogName System -Newest 10</code> <code>journalctl -n 10</code>
Get-EventLog -LogName Application -Newest 5	journalctl -u app.service -n 5	View application logs	Example: <code>Get-EventLog -LogName Application -Newest 5</code> <code>journalctl -u app.service -n 5</code>
Get-EventLog -LogName Security	dmesg	Security/system logs	Example: <code>Get-EventLog -LogName Security</code> <code>dmesg</code>

PowerShell Command	Linux Command	Uses	Example
Get-EventLog -List	journalctl --list-boots	List available logs	Example: <code>Get-EventLog -List</code> <code>journalctl --list-boots</code>
Get-EventLog -LogName System -EntryType Error -Newest 5	journalctl -p err -n 5	Show last errors in logs	Example: <code>Get-EventLog -LogName System -EntryType Error -Newest 5</code> <code>journalctl -p err -n 5</code>
Get-EventSubscriber	tail -f /var/log/syslog	Monitor live logs	Example: <code>Get-EventSubscriber</code> <code>tail -f /var/log/syslog</code>

 Variables & Environment

PowerShell Command	Linux Command	Uses	Example
Get-Variable	set	List environment variables	Example: <code>Get-Variable</code> <code>set</code>
\$env:PATH	echo \$PATH	Show PATH variable	Example: <code>\$env:PATH</code> <code>echo \$PATH</code>
Set-Variable test 'hello'	export test=hello	Set variable	Example: <code>Set-Variable test 'hello'</code> <code>export test=hello</code>
Remove-Variable test	unset test	Remove variable	Example: <code>Remove-Variable test</code> <code>unset test</code>

 Data & Text Processing

PowerShell Command	Linux Command	Uses	Example
Select-String 'error' file.txt	grep 'error' file.txt	Search inside a file	Example: <code>Select-String 'error' file.txt</code> <code>grep 'error' file.txt</code>
Compare-Object file1.txt file2.txt	diff file1.txt file2.txt	Compare files	Example: <code>Compare-Object file1.txt file2.txt</code> <code>diff file1.txt file2.txt</code>
Measure-Object -Line -Word -Character file.txt	wc file.txt	Count lines, words, characters	Example: <code>Measure-Object -Line -Word -Character file.txt</code> <code>wc file.txt</code>

PowerShell Command	Linux Command	Uses	Example
Sort-Object	sort	Sort output	Example: Sort-Object sort
Group-Object	uniq -c	Group/aggregate results	Example: Group-Object uniq -c

 Pipelines & Jobs

PowerShell Command	Linux Command	Uses	Example
ForEach-Object { \$_ * 2 }	xargs -l {} expr {} * 2	Loop through pipeline items	Example: 1..5 \ ForEach-Object { \$_ * 2 } seq 1 5 \ xargs -l {} expr {} * 2
Where-Object { \$_ -gt 10 }	grep -E '^[1-9] [0-9]+\$'	Filter pipeline items	Example: 1..20 \ Where-Object { \$_ -gt 10 } seq 1 20 \ grep -E '^[1-9][0-9]+\$'
Start-Job { Get- Process }	ps &	Run background job	Example: Start-Job { Get-Process } ps &
Receive-Job	jobs ; fg %1	Get job output / manage jobs	Example: Receive-Job jobs ; fg %1

 Miscellaneous & Useful

PowerShell Command	Linux Command	Uses	Example
Get-Command	compgen -c	List all available commands	Example: Get-Command compgen -c
Get-Help Get- Process	man ps	Show command help	Example: Get-Help Get- Process man ps
Get-Date	date	Show current date/time	Example: Get-Date date
Get-Date -UFormat %s	date +%s	Show Unix timestamp	Example: Get-Date -UFormat %s date +%s

PowerShell Command	Linux Command	Uses	Example
Start-Sleep - Seconds 5	sleep 5	Pause execution	Example: Start-Sleep - Seconds 5 sleep 5
Get-Random	shuf -n 1	Generate random number/line	Example: Get-Random shuf -n 1 /usr/share/dict/words
Clear-Host	clear	Clear terminal	Example: Clear-Host clear
New-Guid	uuidgen	Generate UUID	Example: New-Guid uuidgen
Get-History	history	Show command history	Example: Get-History history
Clear-History	history -c	Clear command history	Example: Clear-History history -c